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Energy, Exergy, and Environmental Analysis of Organic Rankine Cycle Systems for Industrial Waste Heat Recovery Applications

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Abstract

In the context of energy transition and the search for sustainable industrial solutions, waste heat recovery is a promising strategy to improve energy efficiency and reduce greenhouse gas emissions. This study investigates the integration of Organic Rankine Cycle (ORC) systems for waste heat recovery through a comprehensive 3E (energy, exergy, and environmental) analysis. A Python 3.10-based simulation framework was employed to model ORC performance under varying operating conditions and working fluids. Two case studies were considered: (i) a metallurgical application (specifically, an aluminium production plant) and (ii) two large marine engines (Man S60-MC6 and Wärtsilä 46DF), evaluated in electricity-only and combined heat-and-power (CHP) modes. Results show that neopentane is the optimal fluid for the aluminum industry, achieving 3.5 MW of net power output with zero environmental penalties. For marine engines, efficiency gains reached 7–8% for the Man engine and over 10% for the Wärtsilä engine in electricity mode, with thermal efficiencies exceeding 35% under CHP operation. The study demonstrates the relevance of ORC systems for the energy recovery of waste heat and the integration of sustainable technologies into industrial processes. It helps improve energy efficiency, reduce environmental impact, and support the energy transition by recovering waste heat.

Keywords: waste heat recovery; Organic Rankine Cycle; sustainable energy systems; environmental impact; energy efficiency; exergy analysis; industrial decarbonization; combined heat and power

1. Introduction

In the current context of global energy transition and the pursuit of sustainable industrial practices, controlling energy demand has become one of the most pressing challenges for the coming decades. According to the International Energy Agency (IEA), global primary energy consumption rose by 2.2% in 2024, nearly twice the average growth rate observed over the previous decade [1]. This trend highlights the urgent need for strategies that both reduce energy consumption and optimize resource utilization. Among all sectors, industry accounts for approximately 30% of global final energy consumption, positioning it as a critical lever for reducing greenhouse gas emissions and improving overall energy performance [2].

In France, industry represents about 22% of final energy consumption, with recent estimates by ADEME (2023) indicating that nearly 30–35% of this energy is dissipated as waste heat during transformation and production processes [3]. This untapped resource amounts to roughly 109 TWh/year, including more than 50 TWh available at temperatures above 100 °C. This data reveals a substantial opportunity for energy recovery and

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decarbonization, particularly in sectors with high thermal losses such as metallurgy [4–6], chemical processing [7–10], food processing [11–13], data centers [14–16] and marine propulsion [17–19]. Harnessing this potential could significantly reduce fossil fuel dependency, lower operational costs, and contribute to achieving national and international climate targets [20].

To address this challenge, Organic Rankine Cycles (ORCs) have emerged as a particularly suitable technology for converting low- and medium-temperature heat into useful energy [21]. ORCs use organic working fluids with lower boiling points, unlike conventional steam cycles. This allows efficient recovery from heat sources that were previously considered unusable. This capability opens new pathways for industrial energy efficiency, especially in applications where heat is available at temperatures between 80 °C and 350 °C [22]. ORCs are increasingly recognized for their modularity, scalability, and compatibility with diverse industrial environments, making them a key technology for waste heat valorization and sustainable energy systems [23].

Several recent studies have reinforced the relevance and versatility of Organic Rankine Cycle (ORC) technology for industrial waste heat recovery [24–26]. Vélez et al. [27] conducted a comprehensive review encompassing technical, economic, and market aspects of ORCs for low-temperature heat conversion. The analysis identified ORCs as a key technology for improving industrial energy efficiency. It also emphasized their growing competitiveness over conventional steam cycles, especially in decentralized applications. Dijoux et al. [28] investigated ORC integration within Ocean Thermal Energy Conversion (OTEC) systems, demonstrating that the selection of working fluid and heat exchanger configuration significantly influences the dynamic behavior and controllability of ORC systems. The study highlighted the critical role of fluid properties and component design in ensuring stable operation. This finding opens the door to better integration in isolated energy networks and renewable systems. Valencia et al. [29] performed a comparative thermodynamic and economic assessment of multiple ORC configurations applied to exhaust heat recovery from a 2 MW gas engine. Their findings revealed that a dual-pressure ORC could enhance overall efficiency by approximately 13% and reduce the specific cost of electricity by nearly 9% compared to a simple cycle. These results confirm the value of multi-criteria approaches for optimizing ORC performance in industrial contexts.

Beyond thermodynamic performance, recent ORC studies increasingly recognize that sustainable waste-heat recovery requires an integrated perspective [30,31]. In this context, working-fluid selection, cycle architecture (e.g., recuperated, regenerative, and multi-pressure layouts), and environmental constraints are treated as key criteria. However, many studies still focus primarily on thermal efficiency or economic indicators. This emphasis can obscure critical sustainability factors, such as component-level exergy destruction and environmental impacts, including global warming potential (GWP) and ozone depletion potential (ODP).

In response, energy–exergy–environmental assessment frameworks and advanced exergo-environmental methodologies have gained traction by coupling exergy-based irreversibility quantification with environmental impact accounting, thereby enabling a more transparent identification of where and why environmental burdens arise within the cycle [32,33]. These works consistently indicate that environmental performance is not governed by cycle efficiency alone but depends strongly on fluid properties and regulatory characteristics, all of which shape the trade-offs required to achieve ORC systems that are simultaneously efficient and environmentally responsible.

However, despite the wide use of 3E methodologies, an important practical limitation remains. Many studies analyze ORC systems in a single operating mode, either electricity-only production or CHP operation, without a unified approach that evaluates both modes

under identical assumptions. This prevents consistent comparisons between operating modes and leaves open the question of how the optimal working fluid, component duties, and component-level exergy destruction change when a single ORC system operates across both modes.

From a modelling standpoint, several open and Python-compatible simulation resources already exist. ORCmKit provides an open-source repository for steady-state ORC components and cycle modelling in Python (and other environments) [34]. ORCSim proposes a general Python framework for ORC simulations and supports multiple architecture and solution strategies [35]. More generally, open-source thermodynamic network tools such as TESPy have been used to simulate ORC power plants under stationary conditions [36]. While these tools provide strong foundations for cycle simulation, they are not typically packaged as a unified, decision-oriented 3E workflow that (i) consistently switches between electricity-only and CHP operation and (ii) compares contrasted real-world waste-heat sources under a harmonised set of energy, exergy, and environmental indicators.

To address these needs, the present work proposes a comprehensive 3E multi-criteria assessment framework for ORC waste-heat recovery, implemented in a Python-based simulation workflow developed in this study. The framework computes energy and exergy indicators at component and cycle levels and integrates environmental metrics aligned with working-fluid and system impacts, enabling consistent performance diagnosis. Critically, the modelling approach supports mode-consistent operation, allowing the same ORC architecture to be evaluated in electricity-only mode and in CHP mode under unified assumptions, thereby enabling robust cross-case comparison.

Two contrasting case studies are analysed to demonstrate the approach and highlight cross-sector insights:

- Waste heat recovery in an aluminum production plant for electricity generation, representing a high-temperature, continuous industrial process.
- Recovery of exhaust heat from two marine engines (MAN S60-MC6 and Wärtsilä 46DF) for CHP production, characterized by distinct thermal profiles and operational constraints.

The results confirm that ORC integration, guided by a 3E multi-criteria methodology, can convert thermal losses into substantial energy gains, reduce environmental impact, and support industrial decarbonization objectives. This work offers a comprehensive framework for advancing waste heat recovery strategies in diverse industrial applications, thereby supporting design choices that are simultaneously efficient and environmentally responsible. The remainder of this paper is organized as follows: Section 2 presents the methodology and modeling framework; Section 3 discusses the results and comparative analysis; and Section 4 concludes with implications for industrial energy recovery and future research directions.

2. Methodology

This study employs a multi-criteria 3E (energy, exergy, and environment) approach to evaluate Organic Rankine Cycle (ORC)-based waste-heat recovery systems in industrial applications. The primary objective is to quantify the thermodynamic performance and energy conversion potential of ORC-based systems operating with various low- and medium-temperature heat sources.

The methodology is built on a Python-based numerical model that simulates the thermodynamic behavior of ORC heat recovery cycles. It evaluates system performance through detailed energy and exergy balances for each component. Additionally, the framework enables comparative analysis of multiple working fluids, based on thermodynamic performance, operational stability, and environmental impact.

2.1. Waste Heat Recovery Cycle: ORC-Based Operation

The ORC-based waste heat recovery cycle converts thermal losses from industrial processes into mechanical work and subsequently into electrical power. Its operation is based on four fundamental thermodynamic transformations, each corresponding to a specific stage of the cycle (Figure 1) [37,38]:

- Heating (Isobaric transformation) (4 → 1): the working fluid is pumped to high pressure and heated at constant pressure within the evaporator until it becomes vapor, using an external heat source such as industrial waste heat or other thermal waste.
- Expansion (Isentropic transformation) (1 → 2): the high-pressure vapor expands isentropically within the turbine, converting thermal energy into mechanical work, which is then used to generate electricity through a coupled generator.
- Condensation (Isobaric transformation) (2 → 3): the expanded vapor is directed into the condenser, where it condenses at constant pressure. The rejected heat is either dissipated to a cold sink (e.g., cooling water) or recovered for secondary thermal applications such as heating or preheating.
- Pumping (Isentropic transformation) (3 → 4): the condensed fluid is pressurized and returned to the evaporator at high pressure. During this step, the fluid's enthalpy increases without a significant change in entropy, completing the closed thermodynamic cycle.

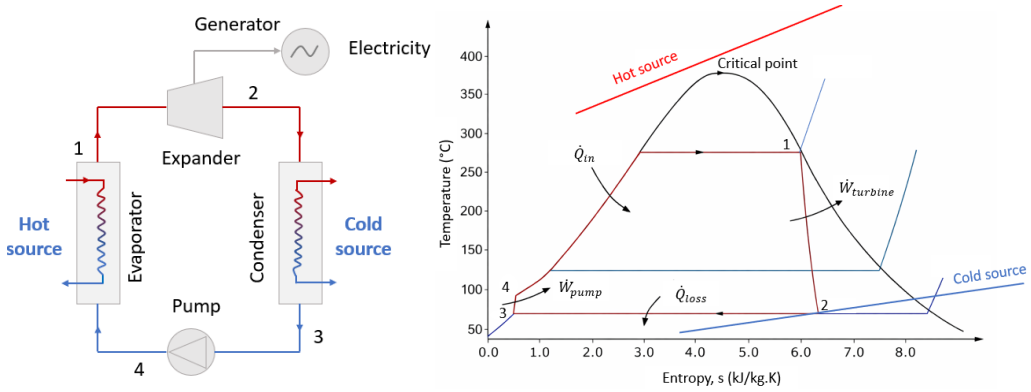


Figure 1. Components of the ORC system (**a**) and T-S diagram of the cycle (**b**).

2.2. Calculation Assumptions

The evaluation of the industrial waste heat recovery cycle relies on simplifying assumptions to streamline numerical modeling:

Steady-state operation: the system operates under steady conditions, and temporal variations in thermodynamic parameters are neglected.

Incompressible working fluid: the fluid is assumed to have negligible volumetric changes under pressure, simplifying the calculation of pumping work.

Simplified cycle configuration: the model includes the main cycle components (evaporator, turbine, condenser, and pump). Parasitic thermal losses are neglected, while fixed pressure drops of 0.5 bar are considered in both the evaporator and the condenser.

These assumptions provide a reliable representation of the cycle's thermodynamic behavior while ensuring the stability and convergence of the calculations.

2.3. Analytical Approaches

Performance is assessed using a multi-criteria 3E framework (energy, exergy, and environment), enabling simultaneous evaluation of system efficiency and sustainability:

Energetic analysis quantifies the cycle's ability to convert supplied heat into useful work, determining overall thermal efficiency.

Exergetic analysis considers both the quantity and quality of energy, accounting for irreversibilities within each component to identify potential improvements.

Environmental analysis examines the environmental impact of the working fluids, mainly through two key indicators: Ozone Depletion Potential (ODP) and Global Warming Potential (GWP).

Integrating these three approaches provides a holistic evaluation of ORC systems, balancing energy performance, thermodynamic efficiency, and environmental sustainability.

2.4. Thermodynamic Modeling

The thermodynamic modeling of the waste heat recovery cycle is based on the first and second laws of thermodynamics, enabling the quantification of energy and exergy performance. The energy balance equations for each component are summarized in Table 1, forming the basis for calculating heat transfer, mechanical work, and net power output.

Table 1. Fundamental energy balance equations of the ORC system [39].

Designation	Balance Equation	Description	
$\dot{Q}_{évap}$	$\dot{m}(h_1 - h_4) = \eta_f \times \sum_{i=0}^{n} \dot{Q}_{Hi}$	Heat supplied to the evaporator	(1)
$\dot{Q}_{cond}$	$\dot{m}(h_2 - h_3)$	Heat rejected by condenser	(2)
$\dot{W}_p$	$\dot{m}(h_4 - h_3)/\eta_p$	Pump power consumption	(3)
$\dot{W}_t$	$\dot{m}\eta_t(h_1 - h_2)$	Turbine power output	(4)
$\dot{W}_{cw}$	$\dot{m}_{cw}(\Delta P_c/\rho_{cw})$	Cooling water circulation	(5)
$\dot{W}_{cy}$	$\dot{W}_t - \dot{W}_p - \dot{W}_{cw}$	Net cycle power	(6)
$\dot{W}_e$	$\eta_g \times \dot{W}_t$	Electrical power generated	(7)

where

$\dot{m}$ is the mass flow rate (kg·s^{-1});

h_1, h_2, h_3 *et* h_4 are the specific enthalpies at each state (kJ·kg^{-1});

$\dot{Q}_{Hi}$ denotes the available thermal power of each hot stream;

$\dot{m}_{cw}$, ρ_{cw} denote the mass flow rate and density of the cooling water, respectively;

ΔP_c represents the pressure losses in the cooling circuit, estimated at 0.5 bar;

η_f, η_p, η_t, η_g correspond to the thermal loss factor and the efficiencies of pump, turbine, and generator, respectively.

2.4.1. Energy and Exergy Efficiencies

The energy efficiency and exergy efficiency of the cycle are defined as [36]:

$$\eta_{en} = \frac{\dot{W}_{cy}}{\sum_{i=0}^{n} \dot{Q}_{Hi}} \tag{8}$$

$$\eta_{ex} = \frac{\dot{W}_{cy}}{\dot{EX}_{Q,Hi}} = \frac{\dot{W}_{cy}}{\sum_{i=0}^{n} \dot{Q}_{Hi}\left(1 - \frac{T_0}{T_{ML,i}}\right)} \tag{9}$$

where $\dot{EX}_{Q,Hi}$, $\dot{Q}_{Hi}$, $T_{ML,i}$ denote, respectively, exergy, available thermal power, and logarithmic mean temperature of each heat source. T_0 the dead-state temperature, which corresponds to the minimum cooling-water temperature (taken as 10 °C).

2.4.2. Cogeneration Mode (CHP Integration)

In addition to electricity generation, waste heat recovery systems can be configured for combined heat and power (CHP) production. CHP integration increases overall heat utilization by recovering the residual thermal energy that is not converted into mechanical power by the ORC [40]. This approach significantly improves overall system efficiency and supports industrial decarbonization by providing both electrical power and useful heat for secondary applications such as process heating or district heating [41]. The following formulation introduces the thermodynamic principles and performance indicators for CHP operation.

To incorporate useful heat recovery in the performance assessment, the exergy efficiency is extended to include the thermal power supplied to the heat user. This conversion is modeled using the Lorentz cycle, which represents the theoretical maximum efficiency for heat transfer between two temperature levels. The corresponding thermal efficiency, denoted as η_{HU}, is expressed as:

$$\eta_{HU} = 1 - \frac{T_0}{T_{ML,HU}} \tag{10}$$

where $T_{ML,HU}$ is the logarithmic mean temperature of the hot water circuit.

The useful heat rate $\dot{Q}_{HU}$ becomes an additional system output, making the thermal efficiency η_{th} and the corrected exergy efficiency relevant indicators for CHP performance.

The thermal and exergetic efficiencies in CHP mode are defined as:

$$\eta_{th} = \frac{\dot{Q}_{HU}}{\sum_{i=0}^{n} \dot{Q}_{Hi}} = \frac{\dot{m}_{HU} C_p \Delta T_{HU}}{\sum_{i=0}^{n} \dot{Q}_{Hi}} \tag{11}$$

$$\eta_{ex} = \frac{\dot{W}_{cy} + \dot{Q}_{HU} \eta_{HU}}{\sum_{i=0}^{n} \dot{EX}_{Q,Hi}} \tag{12}$$

This formulation provides a comprehensive assessment of CHP integration, capturing both mechanical and thermal contributions to the system's overall performance.

To evaluate the overall benefit of waste heat recovery, the performance of the ORC must be considered in conjunction with the base system. For this purpose, global energy, exergy, and thermal efficiencies of the integrated configuration (ORC + Sys) are defined as follows [42]:

$$\eta_{en}(\text{ORC} + \text{Sys}) = \frac{\dot{W}_{cy} + \dot{W}_{Sys}}{\dot{m}_{FUEL} \times LHV} \tag{13}$$

$$\eta_{ex}(\text{ORC} + \text{Sys}) = \frac{\dot{W}_{cy} + \dot{EX}_{Q,HU} + \dot{W}_{Sys}}{\dot{W}_{engine} \times \eta_{com}} \tag{14}$$

$$\eta_{th}(\text{ORC} + \text{Sys}) = \frac{\dot{Q}_{HU}}{\dot{m}_{FUEL} \times LHV} \tag{15}$$

where $\dot{W}_{Sys}$ is the net power of the reference system without waste heat recovery, $\dot{m}_{FUEL}$ and LHV are the fuel mass flow rate and its lower heating value, and $\dot{W}_{engine}$ and η_{com} denote the output power and the combustion efficiency of the internal combustion engine, respectively.

2.5. Numerical Simulation Tool

The performance of the waste heat recovery cycles was evaluated using a numerical simulation tool developed in Python. Figure 2 presents the flowchart describing the

architecture and calculation procedure of this tool, which solves the thermodynamic balance equations introduced in Section 2.4. It also computes the net power output and the energy, exergy, and thermal efficiencies under various operating conditions.

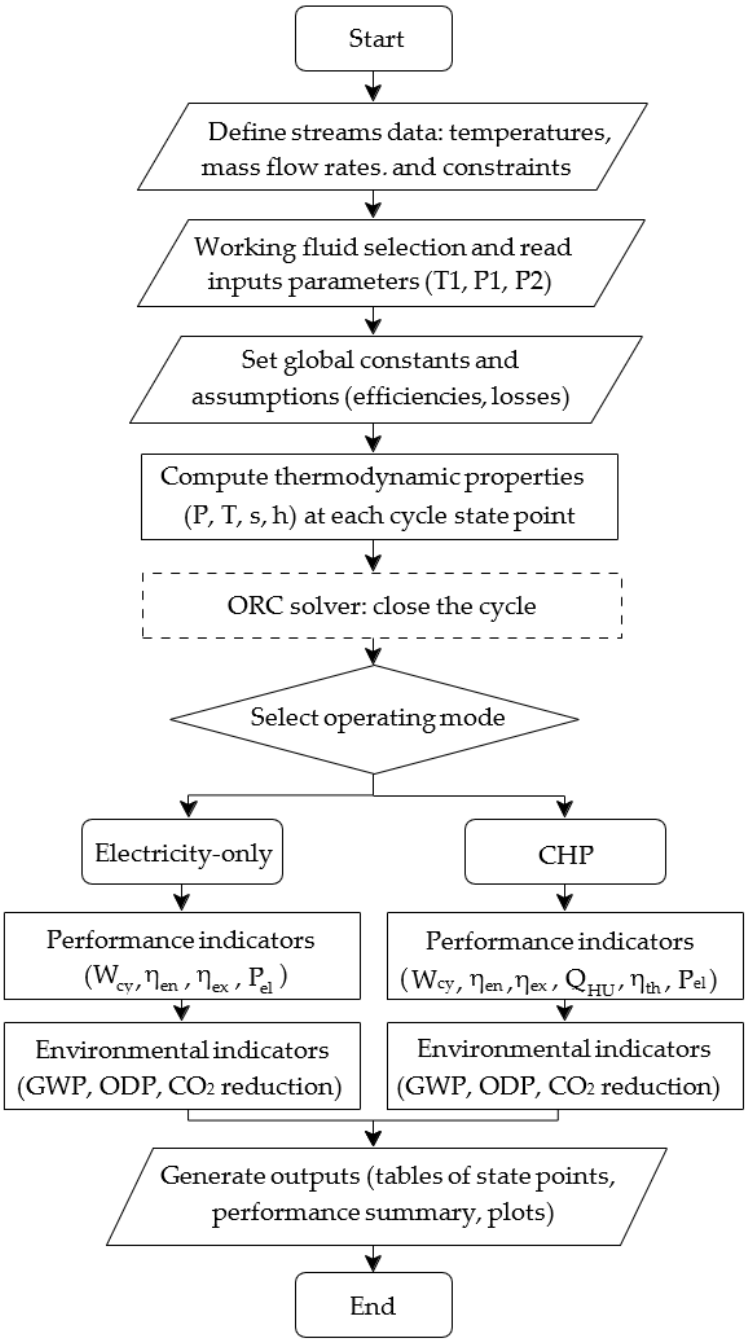


Figure 2. Computational workflow of the numerical tool.

Thermophysical properties of the working fluids were obtained using CoolProp library. Using PropsSI, the thermodynamic state at each cycle point was computed (pressure, temperature, specific enthalpy, entropy, and vapour quality). These calculations depend on the working fluid, the evaporator and condenser boundary conditions, and the isentropic efficiencies of the pump and turbine. An example of the thermodynamic states computed for Propyne is shown in Table 2.

Table 2. Thermodynamic properties at each stage of the ORC cycle using propyne.

Fluid: Propyne						
Index	**Stage**	**P (bar)**	**T (°C)**	**H (kJ/kg)**	**S (J/kg·K)**	**X (%)**
0	1	62.29	172.48	893.10	2833.33	100% Vapor
1	2	5.74	37.64	744.34	2833.33	100% Vapor
2	3	5.74	24.64	254.52	1189.35	100% Liquid
3	4	62.29	27.75	263.73	1189.35	100% Vapor

The simulation tool solves the full set of equations to predict the performance of the waste heat recovery cycle for the two configurations considered: electricity production and combined heat and power (CHP). It also estimates the corresponding efficiency gains achieved through waste heat recovery.

The accuracy of the model was verified using a relative error criterion, defined as:

$$Relative\ error(\%) = \frac{\left|X_{sim} - X_{ref}\right|}{\left|X_{ref}\right|} \times 100 \quad (16)$$

where X_{sim} is the value predicted by the simulation tool and X_{ref} the corresponding reference. This validation step ensures that the predicted performances of the ORC and CHP systems are reliable.

3. Results and Discussion

This section presents the results of the numerical simulations and assesses the performance of ORC-based waste-heat recovery systems under two configurations: electricity-only generation and combined heat and power (CHP). The discussion emphasizes thermodynamic indicators, environmental considerations, and industrial implications to quantify the benefits of waste heat valorization.

3.1. Waste Heat Recovery for Power Generation

3.1.1. Case Study Description—Aluminium Plant

The case study focuses on an aluminum production plant operating electrolytic reduction cell at 940–980 °C, with a specific electricity consumption of approximately 10 kWh/kg of aluminum. During operation, 35% of the electrical input is lost as thermal energy, primarily through combustion gases and shell-cooling air [43]. These streams represent a significant opportunity for energy recovery and efficiency improvement (Figure 3).

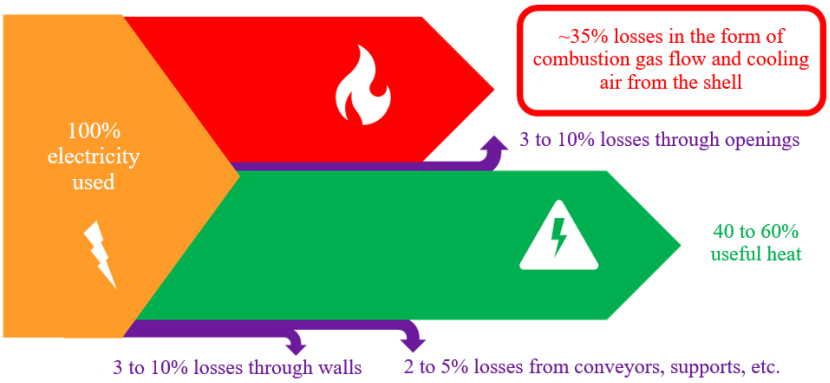


Figure 3. Thermal balance of the furnaces in the aluminum production plant.

The proposed ORC system captures heat from combustion gases (180–80 °C) and shell-cooling air (250–20 °C), with corresponding heat-capacity flow rates of 78 kW/K

and 33.91 kW/K, respectively, as reported in Figure 4. This approach leverages otherwise wasted energy, converting it into mechanical work via a turbine and subsequently into electrical power through a generator. Such integration aligns with decarbonization strategies by reducing primary energy demand and improving the sustainability of aluminum production (Figure 4).

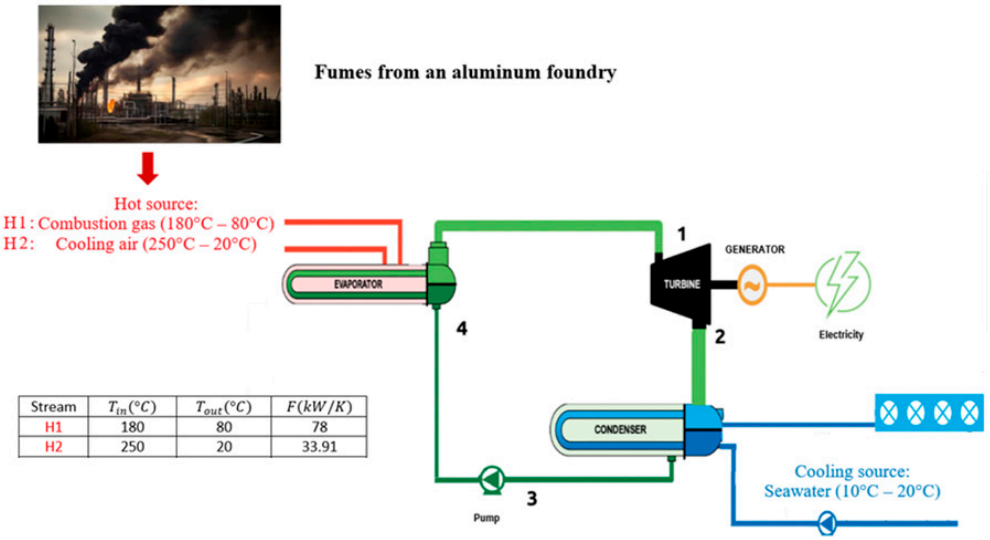


Figure 4. ORC system for waste heat recovery from aluminum foundry fumes.

3.1.2. Model Validation

Before evaluating the system performance, the numerical model was validated using reference data [43]. The calculated evaporator heat input was about 16 MW, which is consistent with the reference operating conditions. As shown in Figure 5, the simulated net power output differs by only 1.8% from the reference values, indicating good accuracy. This small difference is mainly due to the use of different property libraries: CoolProp in this study and RefProp in the reference case. This validation confirms that the model is reliable for further parametric analysis and working fluid selection.

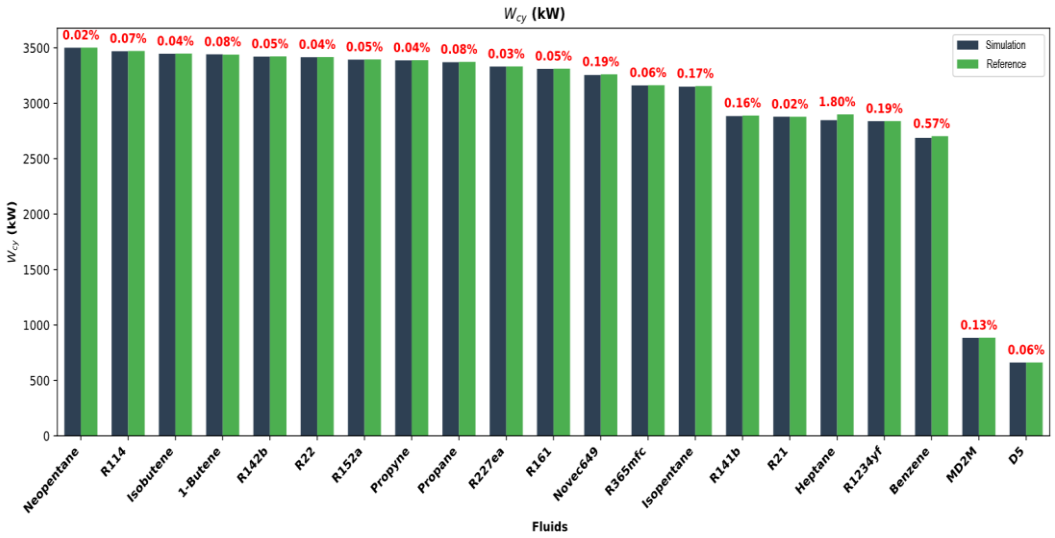


Figure 5. Comparison of simulated and reference values of net power output for different working fluids.

3.1.3. 3E Performance Evaluation for Different Working Fluids

The ORC performance for aluminum foundry was assessed for 21 working fluids, considering:

- Net cycle power (Wcy);
- Energy efficiency (η_{en});
- Exergy efficiency (η_{ex}).

Figure 6 illustrates the comparative thermodynamic performance of the four best-performing working fluids. Neopentane clearly stands out with the highest net cycle power 3.5 MW and exergy efficiency (85.14%), coupled with an energy efficiency of 22.45%, indicating its strong capability to convert low-grade heat into useful work. In comparison, R114, Isobutene, and 1-Butene exhibit competitive results, with net power outputs ranging from 3.41 to 3.46 MW and exergy efficiencies between 83% and 84%. Their energy efficiencies (22.06–22.24%) are slightly lower than Neopentane but remain within a narrow margin, confirming that all four fluids exhibit similar thermodynamic performance, with less than 2% variation in energy efficiency.

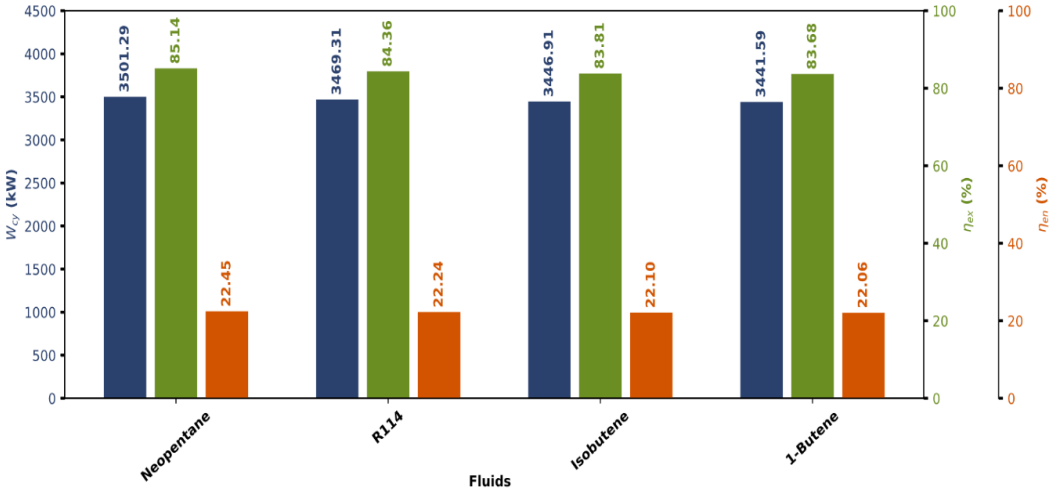


Figure 6. Comparison of the performance of different working fluids.

To complement the energetic and exergetic evaluation, environmental indicators included ODP and GWP were integrated into the analysis, enabling a full 3E assessment (Table 3).

Table 3. Environmental indicators of selected working fluids [43].

Fluid	Neopentane	R114	Isobutene	1-Butene
ODP	0	1	0	0
GWP	0	10,040	0	0

The combined 3E analysis identifies Neopentane as the optimal working fluid, as it combines the highest energy and exergy efficiencies with a net cycle power of 3.5 MW and no environmental impact (ODP = 0, GWP = 0). Although Isobutene and 1-Butene offer thermodynamic performance very similar to R114 (difference of 0.22%), their zero environmental impact makes them more attractive alternatives. Conversely, R114, despite its good thermodynamic performance, has become less suitable due to its high GWP (10,040) and ODP (1).

Beyond these indicators, the ratio between the critical temperature of the working fluid (T_{cr}) and the highest inlet temperature of the heat source (T_{H2}) provides an additional

criterion for fluid selection because it reflects the thermodynamic compatibility between the fluid and the heat source. A fluid with a critical temperature too close to or lower than the heat source temperature may lead to unstable operation or reduced efficiency, while a fluid with a much higher critical temperature can result in poor utilization of the available heat. Therefore, an optimal range of T_{cr}/T_{H2} ensures that the fluid operates in a favorable region of its thermodynamic properties, maximizing exergy efficiency and cycle stability.

As shown in Figure 7a, there is a clear correlation between exergy efficiency and this temperature ratio. Fluids with $0.70 \leq T_{cr}/T_{H2} \leq 0.84$ exhibit the highest exergy efficiencies, and the best-performing fluids in this range correspond to those previously identified (Figure 7b). This correlation confirms the strong influence of the T_{cr}/T_{H2} ratio on exergy efficiency and supports its use as a practical criterion for working fluid selection.

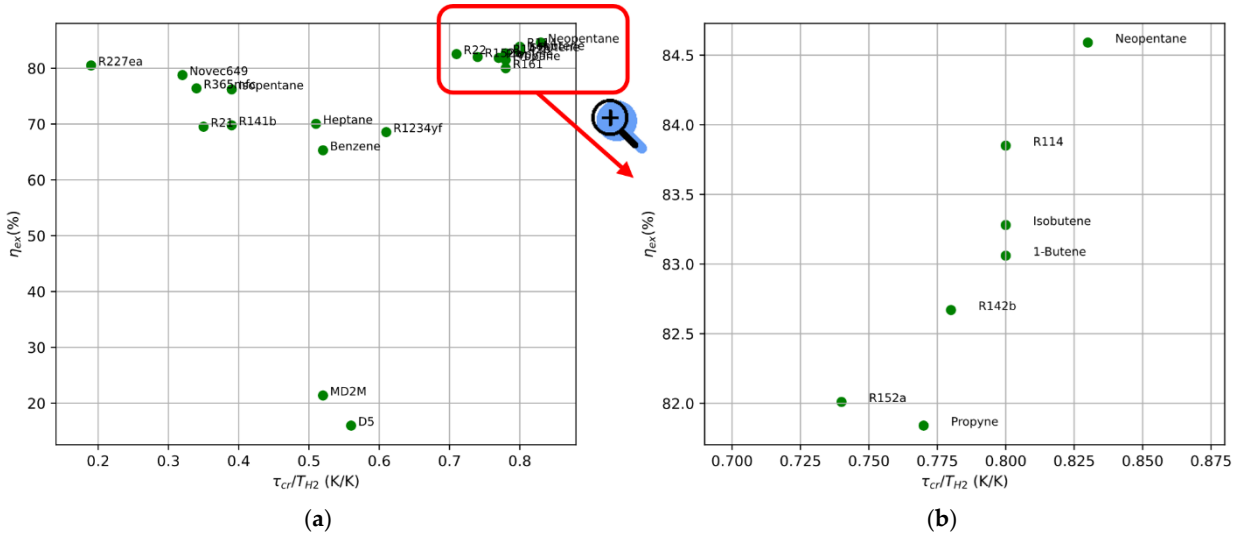


Figure 7. Exergy efficiency as a function of the T_{cr}/T_{H2} (**a**) and best performing fluids (**b**).

Finally, the results of this analysis indicate that Neopentane is the optimal working fluid for maximizing waste heat recovery in the aluminum plant, enabling a net electrical power output of about 3.5 MW. This additional power translates into an increase of approximately 350 kg of aluminum produced per hour and, based on an emission factor of about 10.04 t of CO_2 per tonne of aluminum produced [44], this improvement contributes to a reduction of nearly 3.51 t of CO_2 emissions per hour. This highlights the significant impact of fluid selection on both energy efficiency and environmental performance.

3.2. Waste Heat Recovery for Combined Heat and Power (CHP)

3.2.1. Case Study Description—Combustion Engines

Internal combustion engines, beyond their propulsion function, release substantial thermal energy through high-temperature exhaust gases. These streams represent not only a major source of environmental pollution but also an important reservoir of recoverable energy that is typically lost. This case study examines the potential for recovering and efficiently using the waste heat released from two large marine engines, MAN S60-MC6 and Wärtsilä 46DF, to produce electricity and useful heat in a combined heat and power (CHP) configuration [45].

The hot streams (HS) of these engines include exhaust gas, scavenge air, and jacket water. They are classified by temperature level: HT (High Temperature) for the Man S60-MC6 and VHT (Very High Temperature) for the Wärtsilä 46DF, whose exhaust gases exceed 350 °C and therefore offer greater thermodynamic recovery potential. The seawater circuit

acts as the cold source (CS) for condensation and thermal regulation. Table 4 summarizes the main characteristics of these streams for both internal combustion engines.

Table 4. Main characteristics of internal combustion engines [46,47].

Stream			Man S60-MC6	Wärtsilä 46DF
		Power output	10.3 MW	10.3 MW
		Efficiency	49.59%	45.33%
HS	Exhaust gas	Mass flow rate	26.53 kg/s	19.00 kg/s
		Thermal power	3607 kW	4892 kW
		Temperature range	245–120 °C	354–120 °C
	Scavenge air	Mass flow rate	26.00 kg/s	18.40 kg/s
		Thermal power	3970 kW	3789 kW
		Temperature range	198–48 °C	253–50 °C
	Jacket water	Mass flow rate	21.06 kg/s	23.16 kg/s
		Thermal power	1490 kW	1653 kW
		Temperature range	80–63 °C	91–74 °C
CS	Seawater	Temperature range	10–32 °C	10–32 °C

The ORC-based CHP system is designed to maximize the utilization of waste heat from the engine hot streams, which include exhaust gas, scavenge air, and jacket water. These streams transfer thermal energy to a working fluid circulating in a closed ORC loop. The fluid is vaporized in the evaporator and expanded in a turbine to generate electricity. After expansion, the remaining thermal energy, which cannot be efficiently converted into mechanical power, is recovered during condensation and used to heat a secondary hot water circuit operating between 60 °C and 80 °C, as illustrated in Figure 8. This cogeneration approach significantly improves overall energy efficiency by reducing thermal losses and providing both electrical power and useful heat for onboard or auxiliary applications.

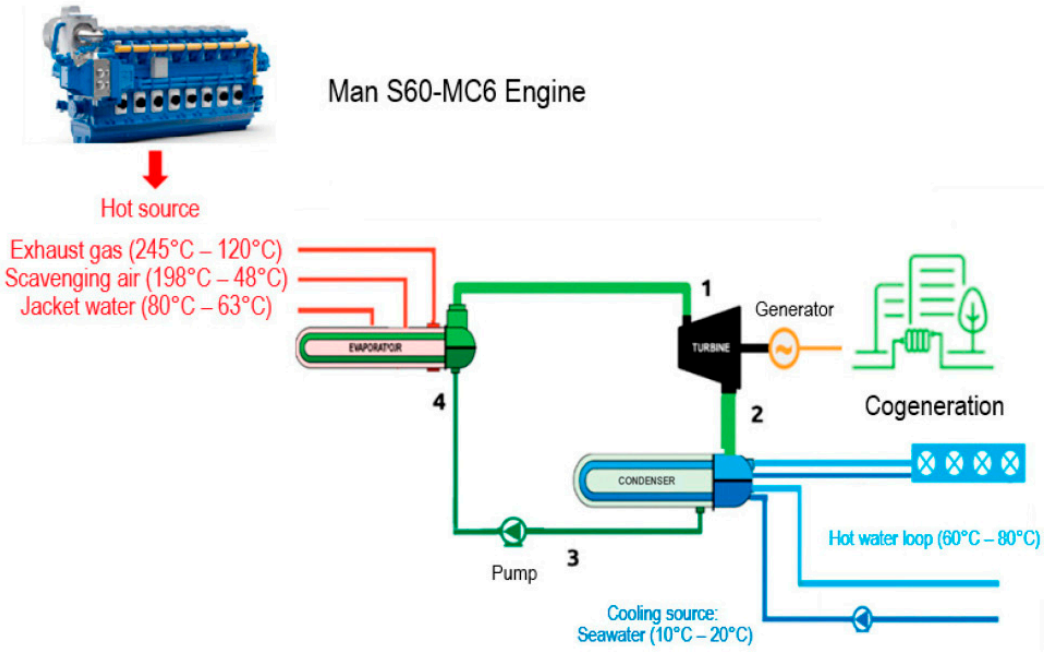


Figure 8. ORC-based CHP system utilizing waste heat from a Man S60-MC6 engine.

3.2.2. Performance Analysis

Before analyzing the cogeneration potential of the engines, the waste heat recovery system was assessed in electricity-only mode. This preliminary evaluation quantifies

the maximum electrical output achievable from the ORC cycle using the thermal energy available in the hot streams of each engine. The analysis considers exhaust gas, scavenge air, and jacket water as heat sources, with seawater as the cooling medium. The results provide a reference for comparing CHP configurations and demonstrate the adaptability of the proposed model to different marine engine profiles.

- Man S60-MC6 Engine

The Python-based simulation framework developed in this study enables the evaluation of both the ORC cycle and the engine's performance after recovering waste heat from the exhaust gases. Figure 9 compares the energy efficiency (η_{en}) and exergy efficiency (η_{ex}) of the ORC, as well as with the overall energy efficiency of the engine after heat recovery $\eta_{en}(ICE + ORC)$ for different working fluids.

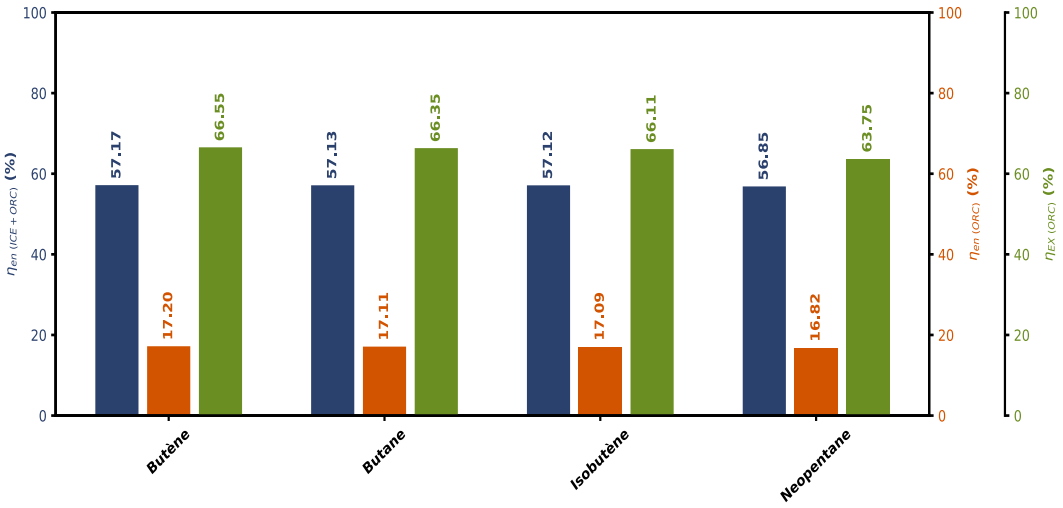


Figure 9. Comparison of performance indicators for electricity production 'Man S60-MC6'.

The results show that using the four selected fluids to convert recovered thermal energy into mechanical power significantly enhances the energetic performance of the Man S60-MC6 engine, whose baseline efficiency is 49.59%. Integrating the ORC increases the overall efficiency by approximately 7–8%, while maintaining a low environmental impact.

Among the fluids considered, butene achieves the most balanced compromise between performance and sustainability. It delivers an additional mechanical power output of about 1.57 MW and reduces CO_2 emissions by nearly 0.83 t per hour, based on an emission factor of 0.53 kg CO_2/kWh [48], while maintaining zero ozone depletion potential (ODP) and a low global warming potential (GWP). These attributes make it slightly preferable to the other fluids evaluated.

For CHP operation, Figure 10 compares the energy efficiency $\eta_{en}(ICE + ORC)$, exergy efficiency $\eta_{ex}(ICE + ORC)$ and thermal efficiency $\eta_{th}(ICE + ORC)$ of the Man S60-MC6 engine for various working fluids, now considering combined heat and electricity production. The 3E analysis shows that all fluids: butene, isobutene, butane, and neopentane enhance overall engine performance, with relatively minor differences among them. The thermal efficiency, reaching approximately 35% for all fluids, represents a major advantage for cogeneration applications, as it enables efficient utilization of recovered heat. Once again, butene stands out slightly due to its superior performance indicators and reduced environmental impact, with zero ODP and low GWP.

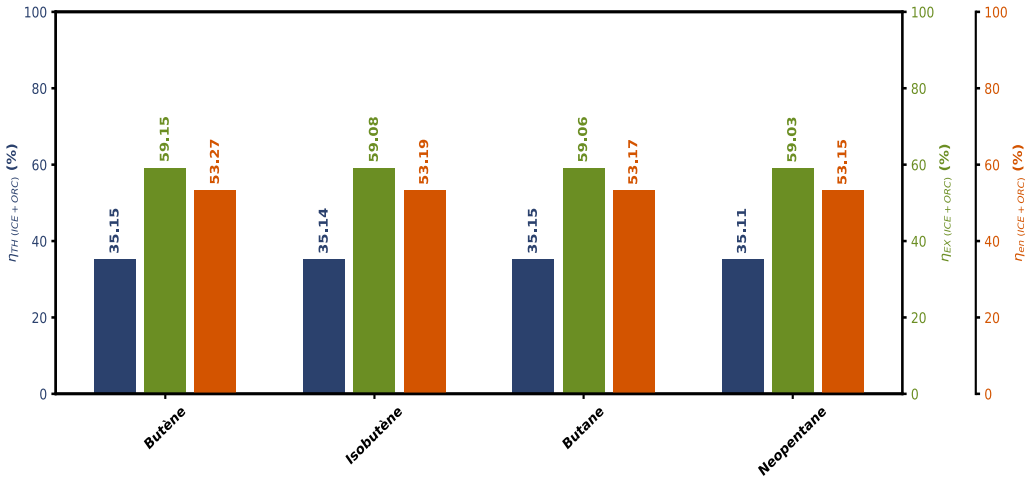


Figure 10. Comparison of Man S60-MC6 engine performance under CHP conditions.

A key benefit of CHP integration is that marine vessels maintain a continuous and diversified demand for thermal energy, arising from essential onboard services such as domestic hot water production, accommodation heating, fuel preheating, lubricating-oil conditioning, and the operation of numerous auxiliary systems. Collectively, these services create a stable thermal base load that can absorb a substantial portion of the recovered heat, ensuring that the CHP system supplies useful onboard energy rather than rejecting this thermal potential to the surrounding environment.

- Wärtsilä 46DF engine

Following the analysis of the Man S60-MC6 engine, the performance of the Wärtsilä 46DF engine was evaluated using the same waste heat recovery approach. The engine operates with higher exhaust gas temperatures, providing more favorable thermodynamic potential for ORC integration. Figure 11 compares the energy efficiency (η_{en}) and exergy efficiency (η_{ex}) of the ORC cycle, as well as the overall energy efficiency of the engine after waste-heat recovery $\eta_{en}(ICE + ORC)$, for different working fluids.

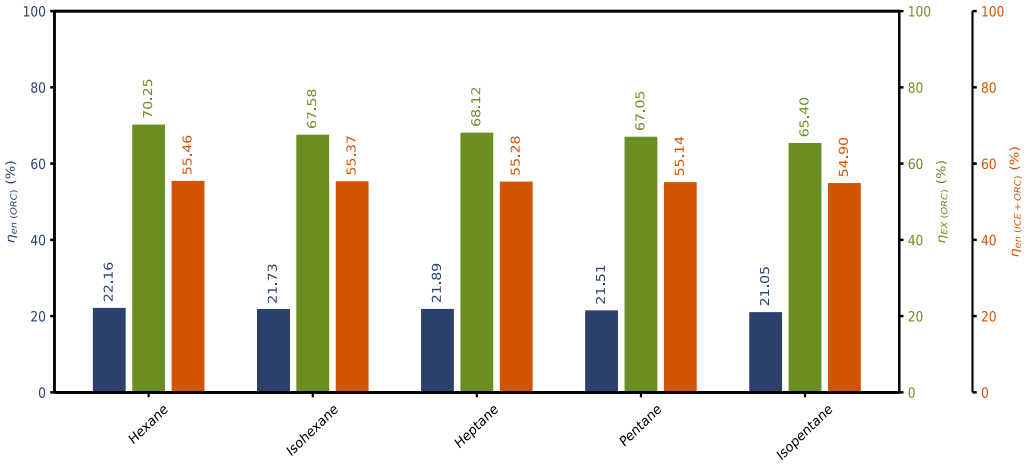


Figure 11. Comparison of performance indicators for electricity production: Wärtsilä 46DF.

The 3E analysis indicates that hexane delivers the highest energetic and exergetic performance within the ORC cycle, increasing the efficiency of the Wärtsilä 46DF engine from 45.33% to 55.46%, corresponding to a significant gain of +10.13%. This efficiency improvement translates into an additional mechanical power output of about 2.29 MW and leads to a reduction of nearly 1.21 t of CO_2 per hour.

However, despite its zero Ozone Depletion Potential (ODP), hexane presents a relatively high Global Warming Potential (GWP) of 14, which limits its environmental appeal. Its higher efficiency gain leads to a larger power increase and, consequently, to a higher reduction in CO_2 emissions compared to the other fluids. Isopentane, although slightly less efficient than hexane and associated with a lower CO_2 reduction, offers a more balanced compromise due to its much lower GWP of 4 and zero ODP, making it the more environmentally favorable option. Thus, while hexane maximizes energetic benefits and CO_2 emission reduction, isopentane remains the preferred fluid when overall environmental criteria are considered, given the relatively small performance difference between the two.

For CHP operation, Figure 12 compares the energy efficiency $\eta_{en}(ICE + ORC)$, exergy efficiency $\eta_{ex}(ICE + ORC)$, and thermal efficiency $\eta_{th}(ICE + ORC)$ of the Wartsila 46DF engine after waste heat recovery for the same set of working fluids. The 3E analysis shows that hexane again provides the highest overall performance, with an energetic gain of +5.91% and a thermal efficiency of 36.65%, confirming its strong thermodynamic suitability for cogeneration applications. Nevertheless, its elevated GWP remains a notable disadvantage from an environmental standpoint. Isopentane, while slightly less efficient, stands out for its significantly lower environmental impact, with a GWP of only 4, which makes it an attractive alternative when prioritizing sustainability. Consequently, the choice between hexane and isopentane depends on whether maximum performance or reduced environmental footprint is considered the primary objective, especially given the marginal difference in their energetic results.

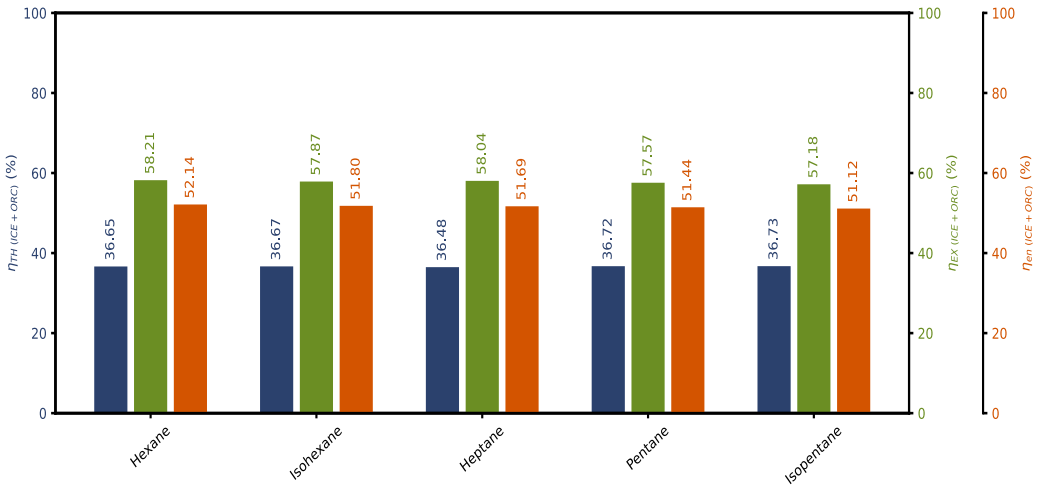


Figure 12. Comparison of Wärtsilä 46DF engine performance under CHP conditions.

The results confirm that both engines benefit significantly from waste heat recovery, with overall energy efficiency improvements ranging from 7–8% for the Man S60-MC6 to over 10% for the Wärtsilä 46DF in electricity-only mode. The higher exhaust gas temperature of the Wärtsilä engine provides a clear thermodynamic advantage, enabling greater ORC performance and higher thermal efficiency under CHP conditions. However, environmental considerations influence fluid selection: while hexane maximizes energetic gains for the Wärtsilä engine, its high GWP makes isopentane a more sustainable choice. For the Man engine, butene offers the best compromise between performance and environmental impact. These findings underscore the importance of balancing technical and ecological criteria when designing ORC-based CHP systems for marine applications.

4. Conclusions

This study presented a comprehensive 3E (energy, exergy, and environmental) assessment of Organic Rankine Cycle (ORC) systems applied to industrial waste heat recovery, using two representative case studies: an aluminum production plant and two large marine engines. A Python-based thermodynamic simulation framework, developed with the CoolProp library, enabled accurate prediction of ORC performance under diverse operating conditions and for a wide range of working fluids.

The first case study demonstrated the potential of ORC integration for electricity generation in the aluminum sector. Among the 21 fluids evaluated, neopentane emerged as the optimal working fluid, delivering the highest net power output (3.5 MW) and the best combination of energy and exergy efficiencies while presenting no environmental penalties (ODP = 0, GWP = 0). The recovered electrical power corresponds to an additional aluminum production rate of approximately 367 kg/h, highlighting the significant industrial and economic benefits of such a recovery system.

The second case study analyzed the valorization of exhaust heat from the Man S60-MC6 and Wärtsilä 46DF marine engines, both in electricity-only and combined heat and power (CHP) modes. For the Man S60-MC6 engine, butene offered the most balanced compromise between energy performance and environmental sustainability, contributing to an overall efficiency increase of around 7–8% and enabling a thermal efficiency of 35% in CHP mode. For the Wärtsilä 46DF engine, hexane achieved the highest energetic and exergetic performance, particularly in CHP operation, while isopentane provided a more environmentally favorable alternative due to its significantly lower GWP, with only a minor reduction in energy performance.

Across both case studies, the 3E multi-criteria approach proved essential for identifying optimal working fluids, revealing that maximizing energy efficiency alone is insufficient without considering exergy degradation and environmental impacts. The results demonstrate that waste heat recovery systems can deliver meaningful improvements in overall system performance, reduce thermal losses, and contribute to industrial decarbonization strategies. The methodology and numerical tools developed in this work provide a robust foundation for further optimization of waste heat recovery technologies in diverse industrial contexts.

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Abbreviations

The following abbreviations are used in this manuscript:

3E	Energy, Exergy, and Environmental
ADEME	French Agency for Ecological Transition

CHP	Combined Heat and Power
CS	Cold Stream
GWP	Global Warming Potential
HS	Hot Stream
HT	High Temperature
IEA	International Energy Agency
LHV	Lower Heating Value
ODP	Ozone Depletion Potential
ORC	Organic Rankine Cycle
T-S	Temperature-Entropy diagram
VHT	Very High Temperature

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